Authentication Page

A REPORT ON

"IIT-B Spoken Tutorial (ST) Courses"

Hosted by: **Department of Computer Science & Engineering.**

Coordinated by: Ms. Ranjitha R

Assistant Professor, Dept of CSE, DBIT, Bengaluru

During: ***September 2025 – January 2026***

Academic Year: **2025 - 2026**

Signature of the Coordinator with date:

Prof.Ranjitha R
IITB_ST Coordinator, Assistant Professor, Dept of CSE, DBIT

Signature of the SPOC with date:

Dr. Venugeetha Y,
IITB_ST SPOC, Professor,
Dept of CSE, DBIT

Signature of the HOD with date:

Signature of the Head-IQAC with date:

Signature of the Principal with date:

PREAMBLE

Spoken Tutorial is a multi-award-winning educational content portal. Here one can learn various Free and Open-Source Software all by oneself. Our self-paced, multilingual courses ensure that anybody with a computer and a desire for learning, can learn from any place, at any time and in a language of their choice.

Courses are simple and easy to follow even for a beginner, but they also meet the growing needs of the learner. Our engaging digital content ensures that learning happens at all levels - Basic, Intermediate and Advanced. Our content mandates side-by-side practice thereby ensuring that learners are actively learning. Many of the software taught, are used in various disciplines of Engineering, pure Sciences and several other Under-Grad and Post- Grad studies, and can be extended to Commerce, Arts and Management streams as well. Alongside these, there are some courses relevant at **School level**, too, which help school students to visualise difficult concepts of Math and Science. These can also be used by Teachers to prepare lesson plans, explain abstract concepts, and give digital assignments to students.

The learning can happen in an organised manner as well. Faculty in institutes can organise their batch of students into groups, who will learn a particular software course for an entire semester. The ST course can be mapped to the Course/Lab manuals and systematic learning can take place. If the academic timetable permits, then a single student can learn up to 3 different ST courses in one semester during the designated academic Lab hours.

End-of-Course online tests and certificates are available for those who wish to test their expertise in a particular software. These certificates give an edge to students during placement by increasing their employability potential.
For more details visit https://spoken-tutorial.org.

Forum help is also available on our website. To post questions on a particular portion of a specific tutorial, visit this web-page: https://forums.spoken-tutorial.org/ .

CIRCULAR

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Department of CS&E (Computer Science and Engineering)

Date: 01/01/2026

CIRCULAR

This is to bring to the notice of all 1st semester ISE students that as part of the IIT Bombay - Spoken Tutorial Program, the following certification courses are offered during the current semester: Python.

The course sessions will be conducted through the official IIT Bombay-Spoken Tutorial platform, and students are expected to register and participate actively in the training modules and complete the assessment.

These courses are aimed at equipping students with essential technical skills and hands-on experience in the respective software/tools, which are widely used in academia and industry. Students are encouraged to make the best use of this opportunity to enhance their technical proficiency beyond their curriculum.

Interested students are informed to register for the above-mentioned courses through the concerned faculty members at the earliest. The schedule for the exam and faculty in-charge are provided in the table below.

Sl.no	Course Name	Date	Faculty Incharge	Signature
1	Python	08/01/2026	Ranjitha R	

Dr. Venugeetha Y
SPOC, IITB

Dr. K B Shiva Kumar
HOD, CSE

Syllabus of PYTHON

1. Getting started with IPython

Outline: >Use Python 3.5.2 >Use Ipython version 5.1.0 >IPython is an enhanced interactive Python interpreter. >Invoke the IPython interpreter >Quit the IPython interpreter >Navigate the IPython session hi..

2. Using plot command interactively

Outline: Use Python 3.4.3 Use Ipython version 5.1.0 Start IPython with pylab. ImportError if matplotlib is not installed clf() function Use the linspace function to create equally spaced points in a regio..

3. Embellishing a plot

Outline: Use Python 3.4.3 Use Ipython version 5.1.0 Modify the attributes of a plot Pass additional keyword arguments to plot command Add title to a plot using 'title' command. Incorporate LaTeX style f..

4. Saving plots

Outline: Use Python 3.4.3 Use Ipython version 5.1.0 Save plots using the savefig() function. Save the plots in different formats like pdf ps png svg eps

5. Multiple plots

Outline: -Use Python 3.4.3 - Use IPython 5.1.0 - Draw multiple plots which are overlaid. - Operations on individual plots. - Use the figure command. - Distinguish between multiple overlaid plots. - Use t..

6. Subplots

Outline: * Use Python 3.4.3 * Use IPython 5.1.0 * Creating subplots * Switching between subplots. * Subplot command * Passing arguments to subplot command. * First argument is the number of rows of subpl..

7. Additional features of IPython

Outline: * Use Python 3.4.3 * Use IPython 5.1.0 * Retrieve the history using %history command. * View only a part of history by passing argument to %history command. * Pass arguments to %history to get pa..

8. Loading Data From Files

Outline: load data from file single column multiple columns separated by delimiter cat command loadtxt() columns separated by spaces columns separated by semi-colon unpack argument delimiter argument ..

9. Plotting Data

Outline: plotting data list list element-wise squaring plot data points clear plots errorbar function dots or filled circles in plot plot with red pluses explore documentation in ipython plot with err..

10. Other Types Of Plots

Outline: scatter plot scatter function scatter plot with various arguments logarithmic plot loglog function cat command loadtxt function unpack parameter of loadtxt linspace scatter versus plot

11. Plotting Charts

Outline: * Use Python 3.4.3 * Use IPython 5.1.0 * To produce scatter plot * Plot a pie chart using pie() function * Plot a bar chart using bar() function * Access the matplotlib online help * Charts with..

12. Getting started with Lists

Outline: What is a list? Define List List index Create: List with elements Empty list List within a list Find out the list length using len function Access elements using their index numbers Appe..

13. Getting started with for

Outline: For loop syntax Example to use For loop Indentation in for loop Create blocks in python using for Iterate over a list using for loop How to get out of the block use of Range() function Range fu..

14. Getting started with strings

Outline: What are strings? How are strings denoted in Python? String concatenation Multiply a string with an integer Accessing individual elements of a string Accessing elements of a string using negative..

15. Getting started with files

Outline: Open a file Open() function Different Modes of opening a file Read() method Read the content of the file line by line Read the entire content of the file Append the lines of a file to a list Cl..

16. Parsing data

Outline: What is Parsing data? split function and its syntax What is string tokenizing? How to split a string on whitespace? split function with argument strip function and example Converting string into..

17. Statistics

Outline: Statistical operations in Python Installation of Numpy for mathematical and logical operations Installation of pip to install python libraries loadtxt() function with example Getting the shape of ..

18. Getting started with arrays

Outline: Overview of array Usage of numpy library How to create arrays How to create two dimensional array arange() method reshape() method How to find the shape of an array? Create a new array with ele..

19. Accessing parts of arrays

Outline: Create a one-dimensional array Create a two-dimensional array Accessing individual elements of an array How to change the value of an array How to change more than one elements at a time Negative..

20. Image manipulation using Arrays

Outline: Read images into arrays How to access parts of an array? imread command imshow command show command How to check the dimensions of an array? Example to access parts of an image How to stride ov..

21. Basic Matrix Operations

Outline: Create matrices from lists asmatrix method arange and reshape methods Basic matrix operation Addition, subtraction and multiplication of a matrix Determinant of a matrix eye(), allclose() functi..

22. Advanced Matrix Operations

Outline: flatten() function Example to convert a multidimensional matrix to single dimension matrix Frobenius norm of a matrix Demonstration of Frobenius norm of a matrix Inverse of a matrix Infinity norm..

23. Least square fit

Outline: Generating a Least Square fit line Generating L vs t square loadtxt function Usage of numpy library Plotting L vs t square Steps for least square fit line Matrix formulation tsq=A.p Generating ..

24. Basic datatypes and operators

Outline: Data types in Python Demonstration of int, float and complex data types with examples Different functions associated with int data type Complex numbers and their functions Boolean operations with ..

25. Sequence datatypes

Outline: List, string and tuple sequence data types with examples How to access a list using index numbers? Access the string elements Access the tuple elements How to add different sequence data types? H..

26. Input-output

Outline: Various output statements Print a string Print a string with newline character How to use format operators? Example for integer format Example for string format Example for float format Getting..

27. Conditional Statements

Outline: if condition statement Demonstration of if statement with example if/else condition statement with example Importance of indentation in a program Usage of colon in program Condition statement usi..

28. Loops

Outline: Explanation of while loop Demonstration of while loop with example Print the squares of all the even numbers below 10 using while loop How to use for loop? Print the squares of all the even number..

29. Manipulating lists

Outline: Various manipulation in lists Slicing of lists Syntax and demonstration of slicing of lists How to use step value in slicing Striding of list Examples with various parameters in striding of list ..

30. Manipulating strings

Outline: How to slice a string Various way to get substrings using index Reverse a string How to check if a given string is a palindrome or not Replace characters in a string Convert a string to uppercase..

31. Getting started with tuples

Outline: What are tuples? How to declare tuples? Examples to declare tuples Demonstration of creating tuple Accessing tuples by their index positions Iteration over tuples Demonstration of immutability p..

32. Dictionaries

Outline: Overview of dictionaries Creating an empty dictionary Creating a non empty dictionary About key:value pair How to access the dictionary elements Demonstration of wrong key Add, delete and modify..

33. Sets in Python

Outline: What are sets in python? Input sets Create sets How to create empty sets? Operations on sets Add and remove methods Union and intersection methods Difference and symmetric_difference methods S..

34. Getting Started with Functions

Outline: About Functions How to define a function Example for defining a function Calling a function with arguments Calling a function without arguments Return values from a function Indentation in codin..

35. Advanced Features of Functions

Outline: Functions with default arguments Various examples for default arguments Interchanging the default and non-default arguments Call a function with keyword arguments Call a function without keyword a..

36. Using Python Modules

Outline: Python modules Run a Python script from command line How to import modules in python scripts? How to import required functions from a module? Usage of namespace Advantages of using import functio..

37. Writing Python Scripts

Outline: About Python modules What is importing? How to write a function and save it as a script Run a python script Import a module Example to import a module usage of __name__ variable Write test cond..

38. Testing and Debugging

Outline: What is software testing? Write a simple function How to write test cases? Create simple tests for a function Run the script and test the code Automate tests Example for test case fail Coding s..

39. Handling Errors and Exceptions

Outline: Errors in Python Syntax errors and exception Exceptions with example Syntax error with example Demonstration of ValueError exception Demonstration of ZeroDivisionError exception try except claus..

Collaborators

IIT Bombay Spoken Tutorial Program,

MoE, Govt of India

In

Association with

DON BOSCO INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course offered by the department along with number of participants

Sl.No	Semester Start Date	Software Course	Department	Participants Strength
1	22/09/2025	Python	ISE	85

Students Registration Details

Participant List_ Python

SL. NO	First Name	Last Name	Signature
1	SHARAT	BG	
2	RAVIKANTH M	M	
3	VIDYASHREE	M	
4	SHREE RAKSHA	LOKESH	
5	RAKSHITA MANJUNATH	RAIKAR	
6	SAGAR B R	.	
7	VIKAS GOWDA	G	
8	VARSHA KK	.	
9	SIDDU	SIDDU	
10	SANKETH	HIREMATH	
11	VIDYASHREE	C	
12	VIKAS	S	
13	RAMYA	SHREE M.S	
14	SHWETHA	SHWETHA	
15	RASHMI	K	
16	RAKESH L R	L R	
17	SHREYANKA P	.	
18	SINDHU.K.V	.	
19	SINDU V	GOWDA	
20	SANTHOSH	B S	
21	SUBRAHMANYA	P BHAT	
22	SAHANA	SAHANA	
23	REETHU	.	
24	TANISH	GOUDAR	
25	VENU	S	
26	SPOORTHI	SHETTY	
27	SHREERAKSHA	MUTTATTTI	
28	THRISHA	P.G	
29	SHRAVANI. K	K	
30	UMA	G	
31	RAMITHA	MANJUNATH	
32	SPOORTHI	N S	
33	VIKASINI	S	
34	SHIVANI.S.L	SHIVANI.S.L	
35	SHALINI	.	
36	YASHMA	CP	
37	SUHASINI S	.	

38	SANDEEP	S	
39	RAKSHITH GOWDA	RAKSHITH	
40	SOWMYA	M	
41	SUGHOSHA K	JOSHI	
42	SHREYA V	SHETTY	
43	ROHITH M	MORE	
44	THEJASHREE ST	.	
45	SAMARTH SURYA	S	
46	TEJAS GOWDA U	.	
47	SUNIL	Y H	
48	LANCHANA N	RAJ	
49	NAVYA	SHREE M	
50	KALYANI	R	
51	K A BHUVAN	KRISHNA	
52	KAVANA	H K	
53	MANOJ	V	
54	RAKESH	HULLOLI	
55	HAYAK	ROSHAN	
56	LAKSHMI A	.	
57	K FURKHAN	AHMED	
58	POORVIKA	MOHAN K	
59	JEEVIKA.	B	
60	MOHAMMAD	FAIZAN	
61	PRASHANTH R	R	
62	NANDHINI	A	
63	PREETHAMSHEKAR	M C	
64	PAVAN R	PAVAN	
65	PUNEETH	KRISHNA S	
66	MIR MOHAMMAD	HYDER	
67	KEERTHANA	R	
68	PRAJWAL	HS	
69	LAVANYA	D	
70	LEELA.	S K	
71	KAUSHIK	POOJARI	
72	NEETA	NAIK	
73	KOMAL GOWDA	K R	
74	KARTIK	VASANT	
75	NAGARAJ	R	
76	NAKUL GOWDA	S	
77	LEAVI	ANDERSON. M	
78	M. SUSHMITHA	.	

79	PAVITHRA	N	
80	POOJA	V	
81	MONALIKA	RATHORE	
82	MANOJ S DODMANI	MANOJ S DODMANI	
83	PUNEETH	GOWDA T	
84	MANASA	.	
85	MALLIKARJUN	CHIMKOD	

BROCHURE


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Utility Software
Git
Version Control System
Python Django
Python Flask
Web Frameworks
CSS
Drupal
HTML
Joomla
Website Building
For more details write to contact@spoken-tutorial.org
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and
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National Mission on Teachers and Teaching (PMMMNMTT), MHRD, Govt. of India

LETTER OF ASSOCIATION

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IIT Bombay

Ref.No. STIITB/2024/5200 Date: - 31/10/2024

LETTER OF ASSOCIATION

To,
The Principal,
Don Bosco Institute Of Technology, Bangalore,
Mysore Road, Kumbalgodu,
Bangalore, Karnataka - 560074.

We are happy to announce the **ASSOCIATION of Knowledge Partner** Spoken Tutorial Program, IIT Bombay with **Don Bosco Institute Of Technology, Bangalore**. Your College is officially now an **Academic Partner of Spoken Tutorial, IIT Bombay**. Spoken Tutorial, IIT Bombay was established with funding from the Ministry of Education, Government of India (previously MHRD), to spread IT Literacy all over India. We are promoting the learning and usage of Free & Open-Source Software (FOSS), through an Audio-Video teaching tool, viz, 'Spoken Tutorial'.

We support and motivate Colleges to train students in Basic Computer, Software and IT Skills. The course and the training are offered for **Rs. 29,500/-** and is valid till **January 2026. This letter is issued from 19th October 2024 to 19th April 2025 to Don Bosco Institute Of Technology, Bangalore and will be renewed after 6 months based on learners trained in the College.**

Looking forward to many enrolments from the College. You are making an outstanding contribution to using ICT-based teaching and learning methodology for students of your College.

For and on behalf of
Spoken Tutorial,
Indian Institute of Technology, Bombay

SPOKEN TUTORIAL PROJECT IIT BOMBAY

Akanksha Saini

Mrs. Akanksha Saini
National Coordinator
Spoken Tutorial Project, IIT Bombay